EXPERIMENT 9

Question: Write the Python program to implement Travelling Salesman Problem.

Aim:

To write a Python program to solve the Travelling Salesman Problem and find the minimum cost tour.

Algorithm:

1. Start the program.
2. Read the distance matrix representing the distances between cities.
3. Generate all possible tours starting from the first city.
4. Calculate the total cost of each tour, including the return to the starting city.
5. Find the tour with the minimum cost.
6. Display the shortest path and minimum cost.
7. Stop the program.

Code:

```
n = int(raw_input("Enter number of nodes: "))

g = []
for i in range(n):
    print "Enter row", i + 1, "of matrix:"
    g.append(map(int, raw_input().split()))

h = map(int, raw_input("Enter heuristic values: ").split())
start = int(raw_input("Enter start node (0-3): "))
goal = int(raw_input("Enter goal node (0-3): "))

cost = [999] * n
parent = [-1] * n
cost[start] = 0
open_list = [start]
closed = []

while open_list:
    u = min(open_list, key=lambda x: cost[x] + h[x])
    open_list.remove(u)

    if u == goal:
        break

    closed.append(u)

    for v in range(n):
        if g[u][v] > 0 and v not in closed:
            new_cost = cost[u] + g[u][v]
            if new_cost < cost[v]:
                cost[v] = new_cost
                parent[v] = u
                if v not in open_list:
                    open_list.append(v)

if cost[goal] == 999:
    print "No path found"
else:
    path = []
    x = goal
```

Output:

```
>>>
Enter number of nodes: 4
Enter row 1 of matrix:
0 1 4 0
Enter row 2 of matrix:
1 0 2 5
Enter row 3 of matrix:
4 2 0 1
Enter row 4 of matrix:
0 5 1 0
Enter heuristic values:  4 2 1 0
Enter start node (0-3): 0
Enter goal node (0-3): 3
Shortest Path: [0, 1, 2, 3]
Total Cost: 4
```

Result:

Thus, the Python program to solve the Travelling Salesman Problem was executed successfully, and the minimum cost tour was obtained.